

KIRS, Russian Federation

WMO: 280090

Lat: 59.350N

Lon: 52.250E

Elev: 554

StdP: 14.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-20.2	-32.0	0.9	-26.7	-25.3	1.4	-20.1	18.3	21.1	16.3	20.2	1.4	160	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	84.4	67.7	80.9	66.0	77.4	63.9	69.7	81.1	67.9	77.5	65.9	74.4	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.1	74.1	64.3	92.0	72.6	62.6	86.7	70.6	34.0	81.2	32.5	77.4	31.0	74.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	13.6	11.7	-34.4	89.1	8.0	2.5	-40.2	90.9	-44.9	92.3	-49.4	93.7	-55.2	95.5		
			-34.6	72.1	8.0	2.2	-40.3	73.7	-45.0	75.0	-49.5	76.2	-55.3	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	8.1	9.8	22.7	37.2	49.6	58.7	63.7	58.0	48.5	36.7	21.7	11.9	(6)
(7)		DBStd	22.06	14.22	13.95	9.29	9.05	9.56	8.01	7.25	7.87	7.49	7.84	11.93	14.86	(7)
(8)		HDD50	6388	1300	1124	845	391	131	15	3	15	116	417	850	1181	(8)
(9)		HDD65	10829	1765	1544	1310	833	482	219	113	246	495	876	1300	1646	(9)
(10)		CDD50	1173	0	0	0	9	119	277	426	264	72	6	0	0	(10)
(11)		CDD65	138	0	0	0	0	5	31	71	30	1	0	0	0	(11)
(12)		CDH74	1514	0	0	0	2	104	351	772	273	12	0	0	0	(12)
(13)		CDH80	368	0	0	0	0	16	74	198	79	1	0	0	0	(13)
(14)	Wind	WSAvg	5.0	5.2	5.2	5.6	5.3	5.4	4.8	3.9	4.2	4.6	5.5	5.3	5.5	(14)
(15)	Precipitation	PrecAvg	23.9	1.5	1.0	1.1	1.4	2.0	2.8	3.1	2.7	2.5	2.5	1.9	1.5	(15)
(16)		PrecMax	34.6	2.7	1.9	2.6	3.4	5.5	8.1	6.6	5.5	4.6	5.9	4.1	2.8	(16)
(17)		PrecMin	14.4	0.3	0.0	0.1	0.4	0.6	0.6	0.7	0.1	0.3	0.3	0.6	0.2	(17)
(18)		PrecStd	3.6	0.7	0.4	0.6	0.8	1.1	1.7	1.4	1.3	1.0	1.1	0.8	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	35.0	47.5	69.9	81.9	86.8	88.3	87.7	75.0	62.5	45.2	38.0	(19)
(20)		MCWB	31.9	32.9	38.4	53.3	62.1	69.3	69.7	69.3	60.9	52.5	43.8	37.3	(20)	
(21)		2%	DB	32.0	33.0	41.7	62.0	76.7	81.9	85.1	81.3	68.5	56.0	40.4	34.4	(21)
(22)		MCWB	30.9	31.6	34.7	49.3	59.0	66.3	68.1	67.3	58.4	48.9	39.1	33.3	(22)	
(23)		5%	DB	29.4	31.1	37.9	56.2	72.1	78.2	81.9	75.9	64.5	51.3	36.5	32.7	(23)
(24)		MCWB	28.3	29.7	33.0	45.8	56.2	63.8	67.1	64.6	56.5	47.1	35.3	31.8	(24)	
(25)		10%	DB	26.4	28.7	35.1	51.4	67.3	74.2	78.4	71.1	60.7	48.0	34.4	30.0	(25)
(26)		MCWB	25.3	27.3	31.9	43.0	53.9	61.9	65.4	62.7	54.9	45.6	33.4	29.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	33.8	39.3	54.4	64.9	71.4	72.0	70.9	63.0	53.7	43.8	37.5	(27)
(28)		MDCB	33.0	34.3	46.4	67.2	78.1	83.6	83.8	83.8	70.6	59.9	45.3	38.0	(28)	
(29)		2%	WB	30.9	31.9	36.2	50.6	60.8	68.4	70.1	68.7	60.1	50.2	39.1	33.6	(29)
(30)		MDCB	31.8	32.8	39.9	60.3	72.8	78.3	81.2	79.0	66.7	54.4	40.2	34.2	(30)	
(31)		5%	WB	28.5	29.8	34.1	47.0	58.2	65.9	68.5	66.1	57.5	47.8	35.6	31.8	(31)
(32)		MDCB	29.4	31.1	36.8	55.1	67.9	74.8	78.7	73.4	62.8	50.7	36.2	32.6	(32)	
(33)		10%	WB	25.3	27.3	32.4	43.7	55.5	63.6	66.8	63.7	55.3	45.5	33.6	29.0	(33)
(34)		MDCB	26.3	28.7	34.8	50.2	65.2	71.9	75.9	70.0	60.2	48.1	34.3	30.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.0	15.3	16.5	17.7	20.4	19.5	20.0	16.7	14.5	9.3	8.2	10.0	(35)	
(36)		5% DB	MCDBR	9.9	11.3	18.5	26.1	28.7	25.6	25.4	22.9	21.3	16.4	7.3	6.8	(36)
(37)		MCWBR	9.8	10.5	13.1	14.9	13.3	11.4	10.6	9.9	12.0	10.7	6.9	7.0	(37)	
(38)		5% WB	MCDBR	9.8	10.2	13.9	24.3	24.1	21.8	22.8	20.8	18.8	14.2	7.1	6.6	(38)
(39)		MCWBR	9.9	10.0	10.4	14.6	12.7	10.9	10.7	10.0	11.5	10.4	6.9	6.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.308	0.354	0.382	0.374	0.367	0.394	0.396	0.359	0.332	0.291	0.258	(40)	
(41)		taud	2.005	2.036	1.985	2.119	2.372	2.427	2.356	2.329	2.407	2.326	2.121	1.973	(41)	
(42)		Ebn at Noon	152	212	238	257	271	275	263	252	244	210	148	112	(42)	
(43)		Edh at Noon	16	27	39	41	34	33	35	33	26	21	15	11	(43)	
(44)	All-Sky Solar Radiation	RadAvg	126	373	748	1202	1574	1697	1708	1240	717	304	114	69	(44)	
(45)		RadStd	18	82	102	163	174	133	173	119	94	37	20	13	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.31	N/A	N/A	N/A	N/A	-493	-589	N/A	N/A
(47)	Regional (6 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	-331	N/A	N/A

Nomenclature: See separate page